

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Haresh Kumar N L	Reg. No.:	192425009
Section / Batch:	AI&ML	Date:	16/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Image Processing System

An image processing algorithm divides images recursively: $T(n) = 4T(n/2) + n$. Apply the Master Theorem to find the time complexity. Clearly identify parameters and case selection. Analyze how performance changes with image size.

Q2. File Sorting in Distributed Systems

A distributed sorting algorithm follows: $T(n) = 2T(n/2) + n \log n$. Apply the Master Theorem and determine the complexity. Justify the selected case. Discuss efficiency in handling large-scale distributed data.

Code of Conduct

/ Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

ANSWER:

Q1. Given: $T(n) = 4T(n/2) + n$

Apply the Master Theorem.

Step 1: Identify Parameters

The standard form is: $T(n) = aT(n/b) + f(n)$

Compare:

$$a = 4, b = 2, f(n) = n$$

Step 2: Compute

$$n^{\log_b a} = n^{\log_2 4} = n^2$$

Since

$$f(n) = n = O(n^{2-1}) = O(n^1)$$

which is polynomially smaller than n^2 ,

$$f(n) = O(n^{\log_b a - \epsilon})$$

where $\epsilon = 1$.

Therefore, Master Theorem Case 1 applies.

Step 3: Time Complexity

$$T(n) = \Theta(n^{\log_2 4}) = \Theta(n^2)$$

Analysis

- The recursive calls dominate the running time.
- As image size increases, execution time grows proportionally to n^2 .
- The algorithm performs well for small images but requires significantly more processing for larger images due to quadratic growth.

Final Answer

- $a = 4$
- $b = 2$
- $f(n) = n$
- Master Theorem Case: Case 1
- Time Complexity: $\boxed{\Theta(n^2)}$

RUBRICS FOR CALCULATION

Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____